

No books, notes, computational devices, etc. are allowed. Use clear handwriting and justify your answers carefully. Please write your anonymous identification number on each sheet of paper.

1. (a) If A is a finite abelian group, show that $\mathbb{Q} \otimes_{\mathbb{Z}} A = 0$.
(b) Show that $\mathbb{Q} \otimes_{\mathbb{Z}} (\mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z}) = 0$.
2. Let G be a group of order 35 and let G act on a set X . Show that if $|X| = 11$, then G must fix at least one point in X .
3. Let G be a group of order 168, and assume that G has a normal subgroup N of prime order. Show that G has a normal Sylow p -subgroup, for some prime p dividing $|G|$.
4. Let R be a ring. An R -module M is said to be *finitely cogenerated* if, for each family $(M_{\alpha})_{\alpha \in \Lambda}$ of submodules of M ,

$$\bigcap_{\alpha \in \Lambda} M_{\alpha} = 0 \implies \bigcap_{\alpha \in \Lambda'} M_{\alpha} = 0$$

for some finite subset Λ' of Λ . Let M be an R -module such that every quotient module of M is finitely cogenerated. Show that every non-empty set S of submodules of M has a minimal element (that is, a submodule M_0 in S such that for any submodule N_0 in S with $N_0 \subseteq M_0$ we have $N_0 = M_0$).

5. Let A be a minimal right ideal in a ring R . Show that either $A^2 = 0$ or $A = eR$, where e is an idempotent in R .
6. Determine the rational canonical form of

$$\begin{pmatrix} 0 & -4 & 85 \\ 1 & 4 & -30 \\ 0 & 0 & 3 \end{pmatrix}.$$